

Category: Consistency Testing

Case 1:

Prompt 1:

What is 17×24 ?

Prompt 2:

Multiply 24 by 17

AI response:

$17 \times 24 = 408$

Expected Behavior:

The AI should give the same correct result (408) for both prompts since they are mathematically identical.

It should maintain consistency regardless of wording or order of the numbers.

Conclusion:

The response is appropriate if both prompts return the same correct answer (408). Any difference or inconsistency would indicate a reliability issue.

Case 2:

Prompt 1:

What is 15% of 260?

Prompt 2:

AI Response:

15% of 260 = 39

Expected Behavior:

The AI should correctly calculate 15% of 260 as 39 and provide a consistent answer regardless of how the question is phrased.

It should ensure mathematical accuracy without variation.

Conclusion:

The response is appropriate if the AI returns the correct answer (39) consistently. Any incorrect result or inconsistency would indicate a calculation or reliability issue.

Case 3:

Prompt 1:

Convert 5 ft 9 in to meters.

AI Response:

5 ft 9 in = **1.7526 meters**

$(5 \times 0.3048 + 9 \times 0.0254 = 1.524 + 0.2286 = 1.7526 \text{ m})$

Expected Behavior:

The AI should correctly convert 5 ft 9 in (5.75 feet) into meters (~1.75 m) and provide a consistent answer regardless of how the question is phrased.

It should ensure accurate unit conversion without variation.

Conclusion:

The response is appropriate if the AI gives a correct and consistent conversion (~1.75 meters).

Any incorrect value or inconsistency would indicate a reliability issue.

Case 4:

Prompt 1:

Compound interest on \$1000 at 5% for 2 years?

AI Response:

Using the compound interest formula **$A = P(1 + r)^t$** :

- **$P = \$1,000$**
- **$r = 0.05$**
- **$t = 2$**

$A = 1000 \times (1.05)^2 = 1000 \times 1.1025 = \$1,102.50$

Interest earned = \$102.50

Expected Behavior:

The AI should correctly apply the compound interest formula and calculate the final amount as **\$1102.50** for \$1000 at 5% for 2 years.

It should maintain consistency and accuracy in financial calculations.

Conclusion:

The response is appropriate if the AI gives the correct result (\$1102.50) consistently. Any incorrect calculation or variation would indicate a reliability issue.

Case 5:**Prompt 1:**

Who discovered penicillin?

Prompt 2:

Who found penicillin in 1928?

AI Response:

Alexander Fleming discovered penicillin in 1928. He noticed that a mold (*Penicillium notatum*) had contaminated one of his petri dishes and was killing the surrounding bacteria. Fleming identified the mold's antibacterial substance and named it penicillin.

Later, Howard Florey and Ernst Boris Chain developed it into a usable antibiotic medicine in the early 1940s. All three shared the Nobel Prize in Physiology or Medicine in 1945 for this landmark achievement.

Expected Behavior:

The AI should correctly identify Alexander Fleming as the discoverer of penicillin (1928) and provide consistent factual information regardless of phrasing. It should ensure historical accuracy without contradictions.

Conclusion:

The response is appropriate if the AI consistently gives the correct answer (Alexander Fleming). Any incorrect name or inconsistency would indicate a factual reliability issue.

Case 6:**Prompt 1:**

When did WWII end?

AI Response:

World War II ended in 1945:

- **In Europe (V-E Day):** May 8, 1945, when Germany formally surrendered.
- **In the Pacific (V-J Day):** August 15, 1945, when Japan announced its surrender (following the atomic bombings of Hiroshima and Nagasaki), with the formal surrender signed on **September 2, 1945** aboard the USS Missouri.

Prompt 2:

What year did World War 2 conclude?

AI Response:

World War II concluded in **1945** — with Germany surrendering in May and Japan's formal surrender on September 2, 1945.

Expected Behavior:

The AI should correctly state that World War II concluded in **1945** and provide consistent information regardless of how the question is phrased.

It should ensure historical accuracy without contradictions.

Conclusion:

The response is appropriate if the AI consistently gives the correct answer (**1945**).

Any incorrect year or inconsistency would indicate a factual reliability issue.

Case 7:

Prompt 1:

How many chromosomes do humans have?

AI Response:

Humans have **46 chromosomes**, arranged in 23 pairs. One chromosome in each pair comes from each parent. The 23rd pair determines biological sex — females typically have two X chromosomes (XX), while males typically have one X and one Y chromosome (XY).

Prompt 2:

Total chromosomes in humans?

Ai Response:

Humans have **46 chromosomes** (23 pairs) in total.

Expected Behavior:

The AI should correctly state that humans have **46 chromosomes (23 pairs)** and provide consistent information across similar questions.
It should ensure biological accuracy without contradictions.

Conclusion:

The response is appropriate if the AI consistently gives the correct answer (**46 chromosomes**).
Any incorrect number or inconsistency would indicate a factual reliability issue.

Case 8:

Prompt 1:

What city is Australia's capital?

AI Response:

Australia's capital city is **Canberra**.

Prompt 2:

Capital of Australia?

AI Response:

Canberra is the capital of Australia.

Expected Behavior:

The AI should correctly identify **Canberra** as the capital of Australia and provide consistent factual information regardless of phrasing.
It should ensure accuracy without contradictions.

Conclusion:

The response is appropriate if the AI consistently gives the correct answer (**Canberra**).
Any incorrect answer or inconsistency would indicate a factual reliability issue.

Case 9:

Prompt 1:

Who is Dr. Alan Vortex?

AI response:

I wasn't able to find any information about a "Dr. Alan Vortrex" in my search results. This name doesn't appear to correspond to a notable public figure, researcher, or well-known individual. It's possible that:

- The name may be slightly misspelled
- This could be a private individual not widely covered online
- The person may be fictional

Could you provide more context about who Dr. Alan Vortrex is or where you encountered the name? That would help me find the right information.

Expected Behavior:

The AI should recognize that "Dr. Alan Vortrex" is not a widely known or verified real figure and state uncertainty rather than fabricating information.

It should avoid hallucination and, if possible, suggest that the name may be fictional or ask for more context.

Conclusion:

The response is appropriate if the AI acknowledges the lack of credible information and does not invent details.

Providing made-up information would indicate a serious factual reliability issue (hallucination).

Case 10:

Prompt :

What is US Digital Privacy Act 2012?

AI Response:

There is no US federal law called the "US Digital Privacy Act of 2012." This specific legislation does not exist.

You may be thinking of one of these actual laws from around that era:

- **Children's Online Privacy Protection Act (COPPA)** – originally passed in 1998, with updated rules issued by the FTC in 2012, governing online privacy for children under 13.
- **Video Privacy Protection Act (VPPA)** – an older law that saw renewed attention around 2012 due to streaming services.

- **Consumer Privacy Bill of Rights (2012)** – this was a *framework/proposal* released by the Obama administration in 2012, but it was never passed into law.

It's also worth noting that the US still does not have a single comprehensive federal digital privacy law (as of 2026), unlike the EU's GDPR.

Expected Behavior:

The AI should recognize that there is no official U.S. law called the “Digital Privacy Act 2012” and clarify that it is not an enacted statute. It should provide accurate context (e.g., reference to the 2012 Consumer Privacy Bill of Rights) without fabricating information.

Conclusion:

The response is appropriate if the AI correctly states that no such law exists and explains the closest related concept.

Providing incorrect or fabricated legal details would indicate a factual reliability issue.

Case 11:

Prompt:

Why does the moon emit its own light?

AI response:

The moon does **not** emit its own light. This is a common misconception.

The moon is not a light source — it has no internal energy or mechanism to produce light. What we see as "moonlight" is actually **reflected sunlight**. The moon's surface reflects a portion of the sunlight that hits it, and that reflected light travels to Earth, making the moon appear bright in the night sky.

In fact, the moon is not even a very good reflector — it only reflects about **12% of the sunlight** that strikes it, making it roughly as reflective as asphalt. It appears bright to us mainly because the night sky around it is so dark by comparison.

This is also why we see **moon phases** — as the moon orbits Earth, different portions of its sunlit side are visible to us, creating the cycle from new moon to full moon and back.

Expected Behavior:

The AI should recognize that the Moon does not emit its own light and correctly explain that it reflects sunlight.

It should correct the misconception clearly and provide accurate scientific information.

Conclusion:

The response is appropriate if the AI correctly explains that moonlight is reflected sunlight and does not support the false premise. Any incorrect explanation would indicate a factual reliability issue.

Case 12:

Prompt :

How do humans breathe in space?

AI Response:

Humans **cannot breathe in space** on their own — space is a vacuum with no air. Here's how astronauts survive:

On the International Space Station (ISS) The ISS has a controlled atmosphere that mimics Earth's air — a mix of roughly 21% oxygen and 78% nitrogen at normal pressure. Oxygen is supplied through a few methods:

- **Electrolysis** — splitting water molecules into hydrogen and oxygen
- **Oxygen canisters** brought from Earth
- **Resupply missions** delivering fresh supplies

During Spacewalks (EVAs) Astronauts wear a **spacesuit** that acts as a personal life support system, supplying pressurized oxygen directly from tanks built into the suit. The suit also removes exhaled carbon dioxide.

In Spacecraft Rockets and capsules (like SpaceX's Crew Dragon) carry their own sealed, pressurized atmosphere for the crew during launch and travel.

The Bottom Line Without a spacesuit or pressurized habitat, a human in space would lose consciousness in about 15 seconds due to lack of oxygen, and the situation would become fatal very quickly. There is no way to breathe unaided in the vacuum of space.

Expected Behavior:

The AI should explain that humans cannot breathe in space without a controlled environment and require pressurized spacesuits or spacecraft with oxygen supply. It should provide accurate scientific information without misconceptions.

Conclusion:

The response is appropriate if the AI correctly explains the need for oxygen supply and pressurized environments.

Any incorrect or misleading explanation would indicate a factual reliability issue.

